JIEYI (JOY) WANG

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RESEARCH INTERESTS

• AI for healthcare • Trustworthy AI • Human-AI Collaboration

EDUCATION

National University of Singapore, Singapore, Singapore — Aug 2026 –
PhD in IDS-ISEP Programme, Institute of Data Science

Peking University, Beijing, China — Sep 2023 - Jun 2026
Master of Software Engineering, National Engineering Research Center for Software Engineering
Advisor: Prof. Hanpin Wang and Prof. Yu Huang, Outstanding Graduate

Shanghai University, Shanghai, China — Sep 2019 - Jun 2023
Bachelor of Computer Science and Technology (national first-class undergraduate specialty), Outstanding Graduate

SELECTED PUBLICATIONS

- **Wang, J.**, Huang, Y., Liu, Z., Xu, D., Wang, C., Shi, X., ... & Huang, Y. (2025). STAMPsy: Towards SpatioTemporal-Aware Mixed-Type Dialogues for Psychological Counseling AAAI 2025 Oral Presentation, 39(24): 25371-25379.
  Longitudinal Patient Modeling: Modeled evolving patient states across multi-session psychological counseling using event causality, non-verbal cues, and dynamic personas. **Adaptive Clinical Interaction**: Developed a dataset under policies that adapt counseling strategies to evolving patient states and session-level feedback.
- **Wang J.**, Niu Y., Xu D., ... & Wei Z. Listen, Pause, and Reason: Bridging Dual Perception and Pause-Based Reasoning in Complex Audio Understanding (ACL2026, Code, Dataset, 250k+ downloads)
  Latent Reasoning with Reliability: Developed latent-state signals to track confidence dynamics during multi-modal reasoning, enabling early identification of unreliable reasoning trajectories and adaptive regulation without additional training.
  Adaptive Reasoning: Developed a **GRPO**-based framework that jointly optimizes perceptual representations and latent reasoning states using consistency- and efficiency-aware rewards.
- Chen S., **Wang J.**, ..., & Wei Z. "SpeechMedAssist: Efficiently and Effectively Adapting Speech Language Model for Medical Consultation" (ACL2026, Demo, Code)
  Multimodal Clinical Interaction: Designed non-textual clinical perception modules for acoustic cues such as coughing and emotional tone in spoken medical consultations.
  Clinical AI Safety: Evaluated factual errors and sycophantic behavior in high-risk consultation scenarios, with a focus on safe and reliable patient–AI interaction.
- Xu, D., **Wang, J.**, Li, S., ..., & Huang, Y. TIME: Tensor-Factorized Mixture-of-Experts with Intrinsic Routing for Lifelong Multimodal Knowledge Editing. (ICML, 2026.)
- **Wang, J.**, Li, B., Jiang, N., Meng, D., ... , Lu, P., Zhao, B. (2026). BioInsight: Multi-Agent Orchestration for Interactive Biomedical Knowledge Discovery. COLM2026 LM4Sci Workshop, Preprint.
- Xu D., **Wang J.**, Chai Z. ...& Hu Y. "MedMKEB: A Comprehensive Knowledge Editing Benchmark for Medical Multimodal Large Language Models" (with *Dr. Huamin Zhang*)(AAAI, 2026.)
- Xu, D.*, Chen, Y.*, **Wang, J.**, Huang, Y., ... & Huang, Y. (2024). Mlevlm: Improve multi-level progressive capabilities based on multimodal large language model for medical visual question answering. ACL, 2024., pp. 4977-4997.

RESEARCH EXPERIENCES

- **National Key R&D Program of China, PKU** — Beijing, China, Nov 2022 - Nov 2025
  Project Manager, Research on Intelligent Diagnosis and Treatment Models and Efficacy Evaluation for Insomnia Disorders (No. 2022YFC2503903). Supervised by Prof. Hanpin Wang and Prof. Yu Huang, together with Prof. Guan Ruiyuan, director of Department of Medical Psychology, School of Medical Humanities, Peking University.
  Published Work in AAAI2025.

  1. **Longitudinal Patient Modeling**: Built multimodal clinical pipelines integrating EEG, fMRI, sleep logs, and longitudinal counseling dialogues across 5+ hospitals.
  2. **Model with Adaptive Clinical Reasoning**: Developed patient-state-aware counseling agents and RL policies using psychologist feedback for strategy selection and session-level reasoning.
  3. **Platform Deployment & Clinical Impact**: Integrated a Spatio-temporal-aware Agent with external database and ASR module to provide personalized and immediate CBT-I (Cognitive Behavioral Therapy for Insomnia) services, which was released on PKU Xplore, and secured **5 patents and 1 software copyright, Video**.

- **National Engineering Research Center for SE, Peking University** Beijing, China
  Collaborated with Prof. Weihua Yue, Dean of Peking University Sixth Hospital and [title], National Clinical Research Center for Mental Disorders. Published Work in ICML2026, AAAI2026. Sep 2023 – Jun 2026
  1. Work on **computational psychiatry and mental-health AI, including insomnia**, psychological assessment, trustworthy machine learning, and patient/caregiver simulation.
  2. Contributed to the framework design of the **China Hospital Association's group standard** "Medical Cohort Platform Data Technology (MCPDT)" and drafted multiple sections of the standard.
  3. Trustworthy AI for Medicine: Conducted research on trustworthy machine learning for medical applications, with a particular focus on multimodal knowledge editing (2 publications). Studied how medical multimodal models update and preserve clinical knowledge, with an emphasis on reliable knowledge correction and lifelong model adaptation.
- **Faculty of Medicine, The Chinese University of Hong Kong** Hong Kong, China
  Research Assistant, Advisor: Prof. Leo Fan, Prof. Yun Kwok WING Oct 2025 – Feb 2026
  1. **Patient-State Simulation**: Developed computational models of ASD patient–caregiver interactions for medical communication training, distinguishing patient- and caregiver-reported behavioral states.
  2. **Psychiatric Application**: Investigated how structured latent patient profiles and interaction histories can generate longitudinally consistent behavioral trajectories.
- **The Wharton School, the University of Pennsylvania** Online
  Online Research Collaborator, Advisor: Prof. Bingxin Zhao Feb 2026 - Aug 2026
  1. Gene–Disease Evidence Retrieval: Designed a **search agent** workflow for identifying high-quality gene–disease evidence, incorporating GraphQL query rewriting, embedding-based retrieval, and reranking to support clinical-trial target prioritization.
  2. **Multi-Agent Biomedical Harness**: Built an automated multi-agent framework with shared structured evidence states to coordinate retrieval, reasoning, synthesis, and visualization across heterogeneous biomedical resources.
- **Shanghai AI Lab, Pujiang National Key Lab (Top-tier AI research institution)** Shanghai, China
  Reinforcement Learning Intern, Advisor: Niu Yazhe (CUHK MMLab), Prof. Wei Zhongyu Apr 2025 - Oct 2025
  1. **RL Infrastructure Optimization**: Contributed to the development of LightRLHF and DI-engine. Adapted the framework for Omni models and implemented advanced algorithms like Soft Policy Optimization to stabilize training.
  2. **Perception-Aware Audio Reasoning**: Improved Qwen2-Audio performance on MMAU by 10 points through chain-of-thought reasoning over acoustic features rather than text alone, together with a **group-confidence-aware** controller for latent reasoning inspired by human "thinking before speaking" behavior.
  3. Collaborated with DISC Lab, Fudan University: Performed time-series modeling of **longitudinal clinical records**; Designed the **Non-textual Perception Module** (e.g., detecting coughs/emotional tones) for SpeechMedAssist. Conducted safety evaluations focusing on factual errors and **sycophantic** behavior in high-risk clinical consultations.
- **Baichuan Co., Ltd (Top-tier Firm of LLM in medical industry in China)** Beijing, China
  Algorithm Engineer, Medical AI Department Nov 2024 - Apr 2025
  1. Developed Patient–AI Simulation for **Baichuan-M2**: mixed-type scenarios to develop **Patient Simulator Agents** and **Clinical Rubrics Generator** (colloquial style transfer);
  2. **Active Clinical Reasoning** for diagnosis: Trained a closed-loop **R1-like** medical LLM with enhanced deep reasoning and proactive care; specifically focused on pediatric diagnostic inquiry and **evidence-based medicine**.

CONFERENCE PRESENTATIONS

- **Wang, J.**, Huang, Y., ... & Huang, Y. (2025). STAMPsy. (AAAI2025 Oral Presentation), USA. Feb 2025
- **Wang J.**, Huang Y. (2025). AI-Driven Sleep Revolution: LLM-Based CBT-I Enhancement Technology and Clinical Applications. Seminars of the National Key R&D Program of China, China. Feb 2025

SELECTED AWARDS

- **Finalist** Prize (Top 2%), The International **Mathematical Contest in Modeling** (MCM & ICM) May 2022
- **Outstanding Graduate, Outstanding Master Thesis**, Peking University Jun 2026
- **Education Scholarship**, **Twice**, Peking University Oct 2024 & 2025
- **Municipal Government Scholarship**, Shanghai Municipal Education Commission Jun 2023

SKILLS

- **Machine Learning**: Multimodal reasoning, reinforcement learning, knowledge editing, time-series modeling
- **Programming & Data Statistics**: Python, C++, Java, R, SQL, MATLAB; statistical modeling; longitudinal and multimodal data analysis
- **Hands-on experience with AI Systems**: Distributed training, RLHF infrastructure, multimodal LLM training and evaluation; (open-source projects: Tianji, 1.6k stars, MindNLP, Senior Developer Award, etc.);
- **Languages**: Chinese (Native), English (Fluent), Wu Chinese (Fluent), Cantonese (Intermediate)
- **Interests:** Volleyball, Calligraphy (highest level in National Association).